Assignment #02

Date: _____

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Computer Organization & Assembly Language

Q 1

```
mov ax, 3
mov bx, ax        ; bx = 3
shl ax, 4         ; 3 x 16 = 48
shl bx, 3         ; 3 x 8 = 24
add ax, bx        ; 48 + 24 = 72

mov bx, 3         ; bx = 3
shl bx, 1         ; 3 x 2 = 6
add ax, bx        ; 72 + 6 = 78
```

Final Answer: ax = 78

Date: تاریخ

Q2

Shift Left by Byte, 4 Times:

```
.data
    value DWORD 2345ABCDh
    lost BYTE 4 DUP(?)
.code
main PROC
mov eax, value
mov ecx, 4
mov esi, OFFSET lost
L1:
main PROC
mov edx, eax
shr edx, 24
mov BYTE PTR [esi], dl

shl eax, 8
inc esi
loop L1
mov value, eax
exit
main ENDP
END main
```

Pg#3

Date:

Shift Right by 1 Byte, 4 times

```
.data
    value DWORD 2345ABCDh
    lost BYTE 4 DUP(?)

.code
    main PROC
        mov eax, value
        mov ecx, 4
        mov esi, OFFSET lost
    L2:
        mov edx, eax
        and edx, 0FFh
        mov BYTE PTR [esi], dl

        shr eax, 8
        inc esi
        loop L2

        mov value, eax
        exit
    main ENDP
END main
```

Date:

Cl 3

Shift Left by 1 word, 2 times

```
.data
    value DWORD 2345A13CDh
    last WORD 2 DUP(?)

.code
    main PROC
    mov eax, value
    mov ecx, 2
    mov esi, OFFSET last
L1:
    mov edx, eax
    shr edx, 16
    mov WORD PTR [esi], dx
    shl eax, 16
    add esi, 2
    loop L1
    mov value, eax
    exit
    main ENDP
END main
```

Date:

Shift Left by 1 Word, 2 times

```
.data
  value DWORD 2345ABCDh
  last WORD 2 DUP(?)
.code
main PROC
mov eax, value
  mov ecx, 2
  mov esi, OFFSET last

L2:
  mov edx, eax
  and edx, 0FFFFh
  mov WORD PTR [esi], dx

  shr eax, 16
  add esi, 2
  loop L2

  mov value, eax
  exit
main ENDP
END main
```

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Q4(a)

```
.data
    BSeconds BYTE ?

.code
main PROC
    mov ax,dx
    mov al, 00011111b
    mov BSeconds, al
    exit
main ENDP
END main
```

Q4(b)

```
.data
    BMinutes BYTE ?
.code
main PROC
    mov ax,dx
    shr ax, 5
    and al, 00111111b
    mov BMintes,al

    exit
main ENDP
END main
```

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CQ5

1) mov dx, 9A3Fh → dx = 9A3Fh

2) mov ax, 7F80h → ax = 7F80h

3) clc → CF = 0

4) add ax, 90EAh → 7F80h + 90EAh = 1106Ah
Therefore, ax = 106Ah, CF = 1

5) adc dx, 1 → 9A3Fh + 0001h = 9A41h
Therefore, dx = 9A41h, CF = 0

6) stc → CF = 1

7) add ax, 8120h → 106Ah + 8120h = 918Ah
Therefore, ax = 918Ah, CF = 0

8) adc dx, 0FFFFh → 9A41h + FFFFh = 19A40h
Therefore, dx = 9A40h, CF = 1

Final Answer:

ax = 918Ah

dx = 9A40h

CF = 1

Date: _____ تاریخ

Q6

a)
```
test ax, ax
```

b)
```
test bx, 1
```

c)
```
test dx, 8000h
```

d)
```
test dx, 8000h
```

e)
```
test bl, bl
```

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CO7

1) mov dx, 5

dx = 5

2) stc

set CF = 1

3) mov ax, 10h

10h is hex for 16 dec

ax = 16

and mov does not change CF so:

CF = 1

4) adc dx, ax

adc adds the destination, source, and current carry Flag.

dx = dx + ax + cf

Substituting:

dx = 5 + 16 + 1 = 22

In hex:

$22_{10} = 16h$

Therefore, dx = 16h

ax = 10h

CF = 0

Date: تاریخ

Q8

```
.data
val1 QWORD
.code
main PROC
    mov esi, OFFSET val1
    mov edi, OFFSET val2
    mov ebx, OFFSET result
    mov ecx, 8
    clc
L1:
    mov al, BYTE PTR [esi]
    sbb al, BYTE PTR [edi]
    mov BYTE PTR [ebx], al
    inc esi
    inc edi
    inc edx
    loop L1
    exit
main ENDP
END main
```

Date:

Q9

```
.data
    prompt BYTE "Enter marks (0-100
: ", 0
    gradeA  BYTE "Grade: A", 0
    gradeB  BYTE "Grade: B", 0
    gradeC  BYTE "Grade: C", 0
    gradeF BYTE "Grade: F", 0

.code
main PROC
    mov edx, OFFSET prompt
    call writeString
    call ReadInt
    cmp eax, 80
    jge AGrade
    cmp eax, 60
    jge BGrade
    cmp eax, 40
    jge CGrade

AGrade:
    mov edx, offset gradeA
    call WriteString
    jmp Done
```

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```
BGrade:
    mov edx, offset gradeB
    call WriteString
    jmp Done

CGrade:
    mov edx, offset gradeC
    call WriteString
    jmp Done

FGrade
    mov edx, offset gradeF
    call writeString
Done:
    call Crlf
    exit
main ENDF
END main
```

Date:

Q10

Instruction	EAx	EBx	Stack
mov eax, 3	3	-	Empty
mov ebx, 7	3	7	Empty
push eax	3	7	3
push ebx	3	7	7, 3
~~push~~ pop eax	7	7	3
push 10	7	7	10, 3
pop ebx	7	10	3
pop eax	3	10	Empty

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Q 11

```
.data
num1   SDWORD 10
num2   SDWORD 5
num3   SDWORD 8
num4   SDWORD 3
result SDWORD ?

.code
main PROC
   mov eax,num1
   cmp eax,num2
   jle CheckElseIf

   mov eax,num3
   cmp eax,num4
   je CheckElseIf

   mov eax,num1
   add eax,num3
   mov result,eax
   jmp Done
```

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```
CheckElseIf:
    mov eax, num2
    cmp eax, num4
    jne Otherwise

    mov eax, num2
    sub eax, num1
    mov result, eax
    jmp Done

Otherwise:
    mov result, 0

Done:
    mov eax, result
    call WriteInt
    call Crlf

    exit
main ENDP
END main
```

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Q12

```
.data
op1 SDWORD 5
op2 SDWORD 10
x   SDWORD 3
y   SDWORD 7
z   SDWORD 0

.code
main PROC
OuterLoop:
    mov eax, op1
    cmp eax, op2
    jg Done

InnerLoop:
    mov eax, op1
    cmp eax, op2
    jg EndInner

    mov eax, op1
    cmp eax, x
    jl ElsePart
```

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```
        cmp eax,y
        jg ElsePart

        add z,10
        jmp InnerLoop

ElsePart:
        sub z,10
        jmp InnerLoop

EndInner
        dec op1
        jmp OuterLoop

Done:
        mov eax,z
        call WriteInt
        call Crlf

        exit
main    ENDP
END main
```